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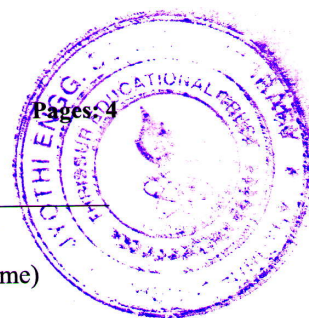
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Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)



Course Code: MAT204

Course Name: PROBABILITY, RANDOM PROCESSES AND NUMERICAL METHODS

Max. Marks: 100

Duration: 3 Hours

(Statistical Tables are allowed)

PART A

(Answer all questions; each question carries 3 marks)

- | | | Marks |
|----|---|-------|
| 1 | In a binomial distribution, if the mean is 6, and the variance is 4, find $P[X=1]$. | 3 |
| 2 | Given that $f(x) = \frac{K}{2^x}$ is a probability mass function of a random variable that can take on the values $x = 0, 1, 2, 3$ and 4, find (i) K and (ii) $P(X \leq 2)$. | 3 |
| 3 | Find the mean and variance for the PDF, $f(x) = \begin{cases} Kx^2, & 0 < X < 1 \\ 0, & \text{elsewhere} \end{cases}$ | 3 |
| 4 | If random variable X has a uniform distribution in $(-3, 3)$, find $P(X - 2 < 2)$. | 3 |
| 5 | Define stationary random process. Define two types of stationary random process. | 3 |
| 6 | Write down the properties of the power spectral density. | 3 |
| 7 | Write down the Newton's forward and backward difference interpolation formula | 3 |
| 8 | Evaluate $\int_0^1 \frac{1}{1+x^2} dx$ with 4 subintervals by Simpson's rule. | 3 |
| 9 | Write the normal equations for fitting a curve of the form $y = a + bx + cx^2$ to a given set of pairs of data points. | 3 |
| 10 | Using Euler's method, find y at $x = 0.25$, given $y' = 2xy$, $y(0) = 1$, $h = 0.25$. | 3 |

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- | | | |
|----|---|---|
| 11 | a) Six dice are thrown 729 times. How many times do you expect at least three dice to show 1 or 2? | 6 |
| | b) Derive the formula for mean and variance of Poisson distribution | 8 |
| 12 | a) A random variable X takes the values -3, -2, -1, 0, 1, 2, 3 such that $P(X=0) = P(X>0) = P(X<0)$ and $P(X=-3) = P(X=-2) = P(X=-1) = P(X=1) = P(X=2) = P(X=3)$. Obtain the probability mass function and distribution function of X. | 7 |

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- b) The joint probability distribution of X and Y is given by, $f(x, y) = \frac{1}{27}(2x + y)$; 7
 $x = 0, 1, 2$ and $y = 0, 1, 2$.
(i) Find the marginal distributions of X and Y.
(ii) Are X and Y independent random variables.

Module -2

- 13 a) Suppose the diameter at breast height (in.) of trees of a certain type is normally distributed with mean 8.8 and standard deviation 2.8. (i) What is the probability that the diameter of a randomly selected tree will be at least 10 in.? (ii) What is the probability that the diameter of a randomly selected tree will exceed 20 in.? (iii) What is the probability that the diameter of a randomly selected tree will be between 5 in. and 10 in.? 7
- b) The amount of time that a surveillance camera will run without having to be reset is a random variable having exponential distribution with mean 50 days. Find the probabilities that such a camera will (a) have to be reset in less than 20 days. (b) not have to be reset in at least 60 days. 7
- 14 a) The joint density function of 2 continuous random variable X and Y is 7
$$f(x, y) = \begin{cases} cxy & ; 0 < x < 4, 1 < y < 5 \\ 0 & ; \text{otherwise} \end{cases}$$

(i) Find the value of the constant c.
(ii) Find $P(X \geq 3, Y \leq 2)$
(iii) Find the marginal density of X.
- b) The life time of a certain brand of tube light may be considered as a random variable with mean 1200 hours and standard deviation 250 hours. Using Central limit theorem, find the probability that the average life time of 60 lights exceeds 1250. 7

Module -3

- 15 a) Let $X(t) = A \cos \lambda t + B \sin \lambda t$, where A and B are independent normally distributed random variables $N(0, \sigma^2)$. Show that X(t) is WSS. 7
- b) If $X(t) = A \cos(\omega t + \theta)$ Where A and ω are constants and θ is uniformly distributed over $[0, 2\pi]$, find the auto correlation function and Power Spectral Density of the process. 7

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- 16 a) Assume that $X(t)$ is a random process defined as follows: $X(t) = A \cos(2\pi t + \phi)$ 7
 where A is a zero-mean normal random variable with variance $\sigma_A^2 = 2$ and ϕ is uniformly distributed random variable over the interval $-\pi \leq \phi \leq \pi$. A and ϕ are statistically independent. Let the random variable Y be defined as $Y = \int_0^1 X(t) dt$. Determine (i) The mean of Y (ii) The variance of Y .
- b) If the customers arrive at a counter in accordance with Poisson distribution with rate 7
 of 2 per minute. Find the probability that the interval between two consecutive arrivals is (i) more than 1 minute (ii) between 1 minute and 2 minutes.

Module -4

- 17 a) Use Newton- Raphson method to find a non- zero solution of $f(x) = 2x - \cos x = 0$ 7
 b) Using Lagrange's interpolating polynomial estimate $y(5)$ for the following data: 7

x	1	3	4	6
y	-3	0	30	132

- 18 a) Find the polynomial interpolating the following data, using Newton's backward 7
 interpolating formula

x	3	4	5	6	7
y	7	11	16	22	29

- b) Using Newton's divided difference formula, evaluate $y(8)$ and $y(15)$ from the 7
 following data

x	4	5	7	10	11	13
y	48	100	294	900	1210	2028

Module -5

- 19 a) Solve the following system of equations using Gauss- Seidel iteration method 7
 starting with the initial approximation $(0,0,0)^T$

$$8x_1 + x_2 + x_3 = 8$$

$$2x_1 + 4x_2 + x_3 = 4$$

$$x_1 + 3x_2 + 5x_3 = 5$$

- b) Fit a straight-line $y = ax + b$ for the following data: 7

X	1	3	4	6	8	9	11	14
Y	1	2	4	4	5	7	8	9

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- 20 a) Solve the following system of equations using Gauss- Jacobi iteration method 7
starting with the initial approximation $(0,0,0)^T$

$$20x_1 + x_2 - 2x_3 = 17$$

$$3x_1 + 20x_2 - x_3 = -18$$

$$2x_1 - 3x_2 + 20x_3 = 25$$

- b) Use Runge - Kutta method of fourth order to find $y(0.1)$ from $\frac{dy}{dx} = \sqrt{x+y}$, 7
 $y(0)=1$ taking $h=0.1$.

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